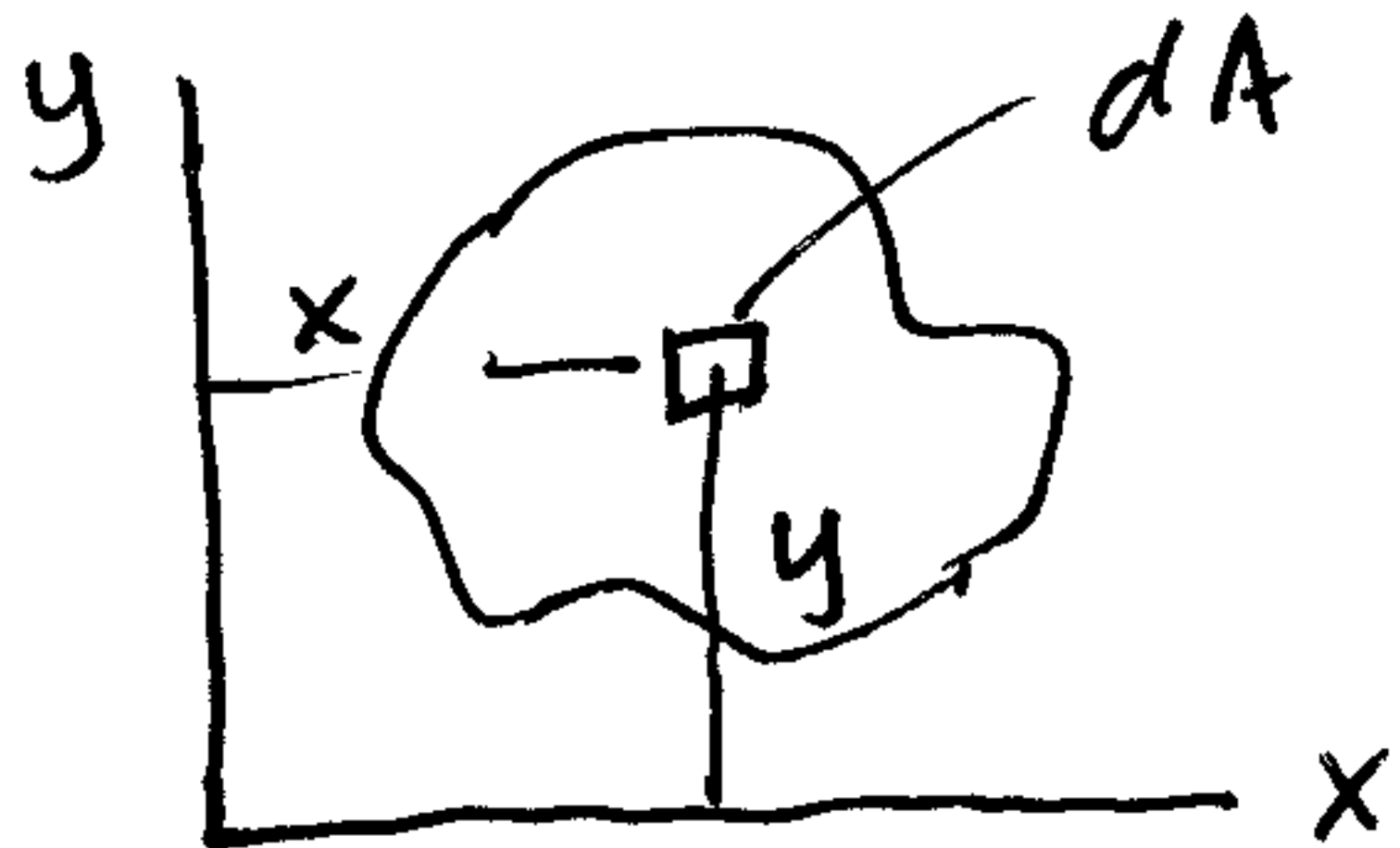


Product of Inertia : Introduction

(I_{xy} or POI) ①

Recall MOI's :



$$I_x = \int_A y^2 dA$$

$$I_y = \int_A x^2 dA$$

- For double integrals
- For properly chosen dA 's ...

I_x : Parallel to x axis

I_y : Parallel to y axis

Product of Inertia (POI) :

$$I_{xy} = \int_A xy dA$$

✓ only really true for double integral. (see above.)

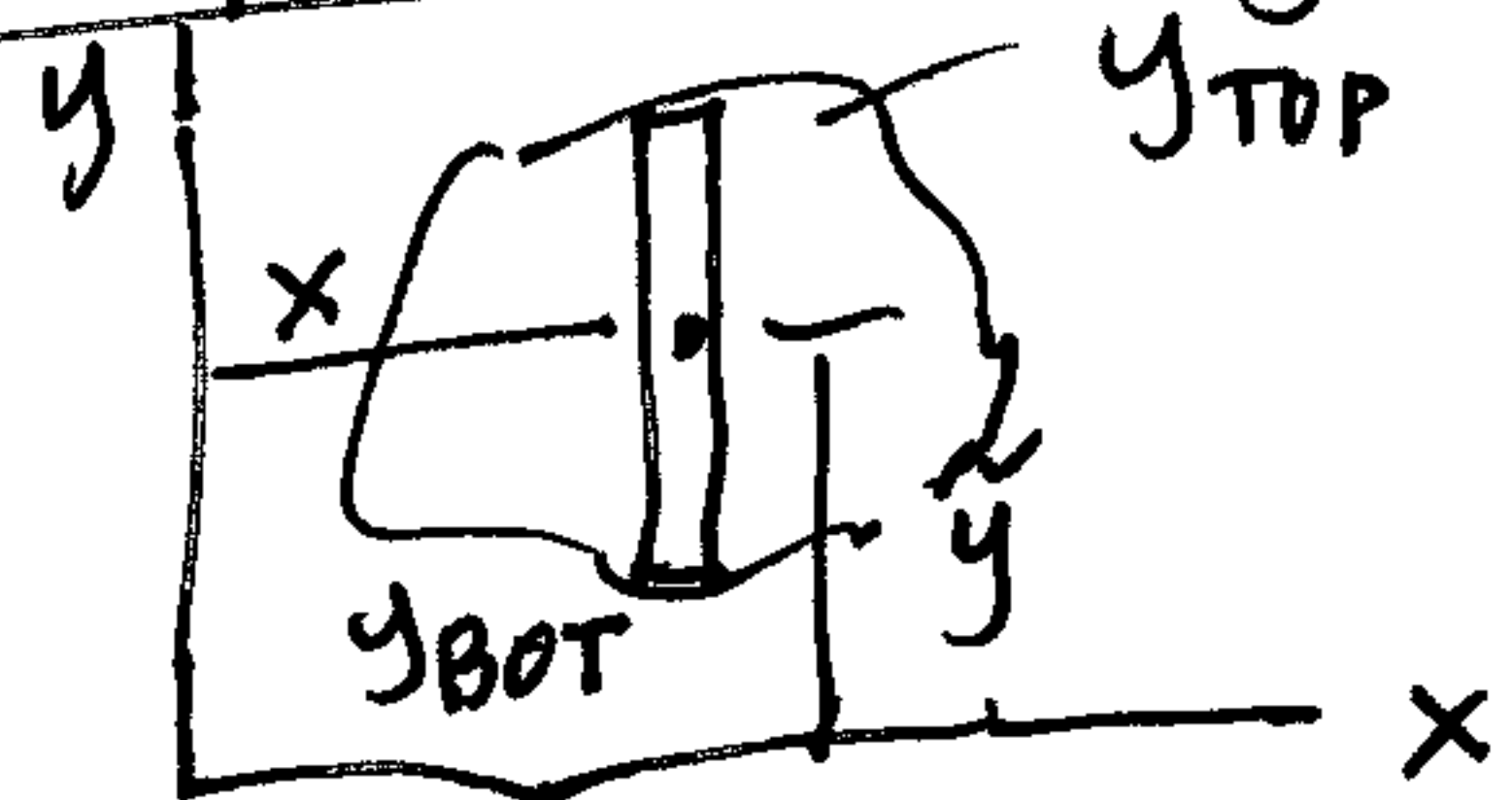
If dA parallel to x :



$$I_{xy} = \int_A \tilde{x} \cdot y dA$$

where, $\tilde{x} = \left(\frac{x_L + x_R}{2} \right)$

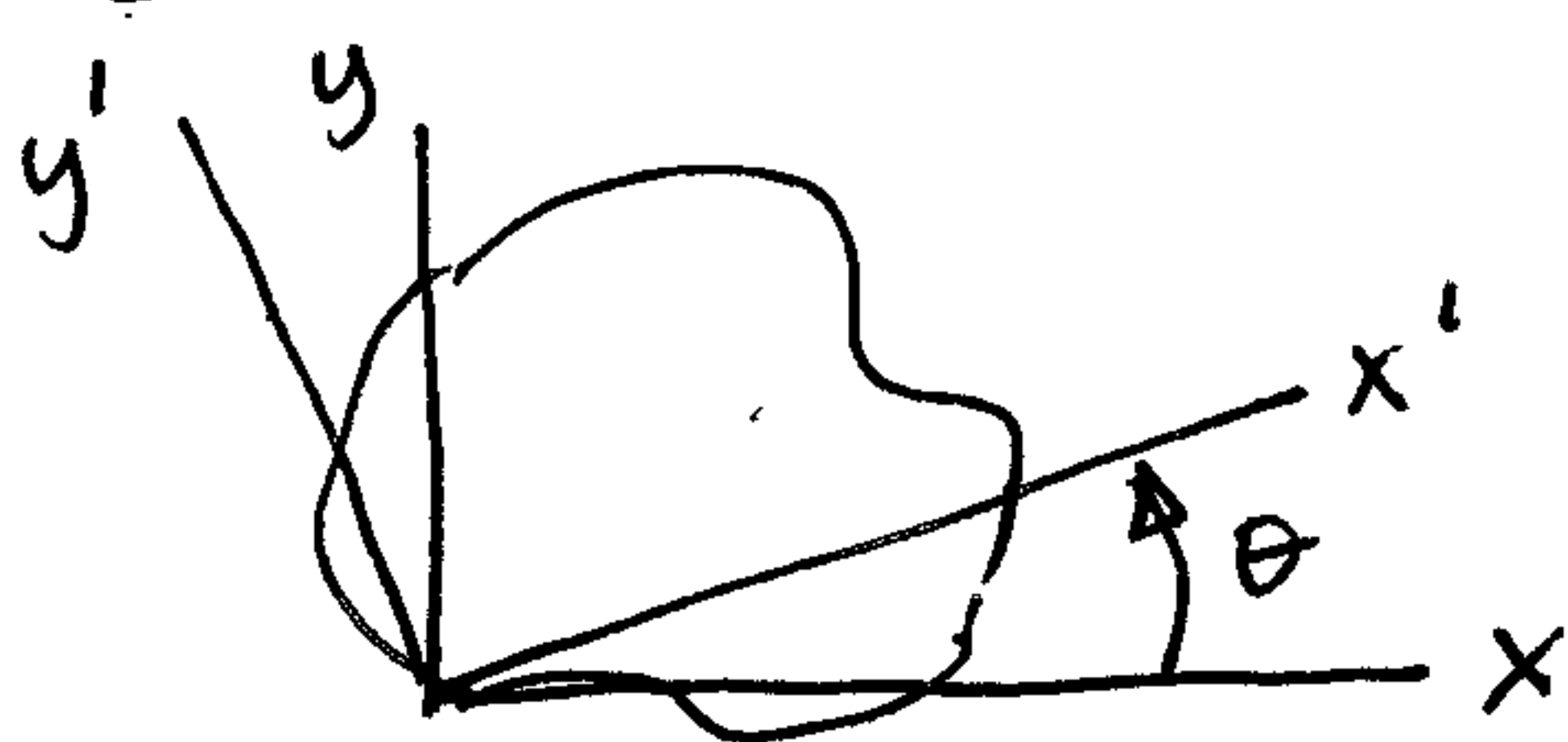
If dA parallel to y :



$$I_{xy} = \int_A x \tilde{y} dA$$

where, $\tilde{y} = \left(\frac{y_{bot} + y_{top}}{2} \right)$

Why need POI ?



We will derive "transformation" equations which, if given I_x , I_y and I_{xy} , can compute

$I_{x'}$, $I_{y'}$, $I_{x'y'}$ at any θ measured from original $y \perp x$.

I_{xy} arises from that derivation.

Need all three (I_x , I_y , I_{xy}) in order to compute the rotated values $I_{x'}$, $I_{y'}$, $I_{x'y'}$.

For a given area, I_x and I_y are always positive ⁽²⁾
and cannot be zero...

because of the x^2 and y^2 terms:

$$I_x = \int_A y^2 dA$$

$$I_y = \int_A x^2 dA$$

However, POI (I_{xy}) can be 0, +, or Negative!

because of the xy term:

$$I_{xy} = \int_A xy dA$$

For a shape w/ an x or y axis of symmetry,

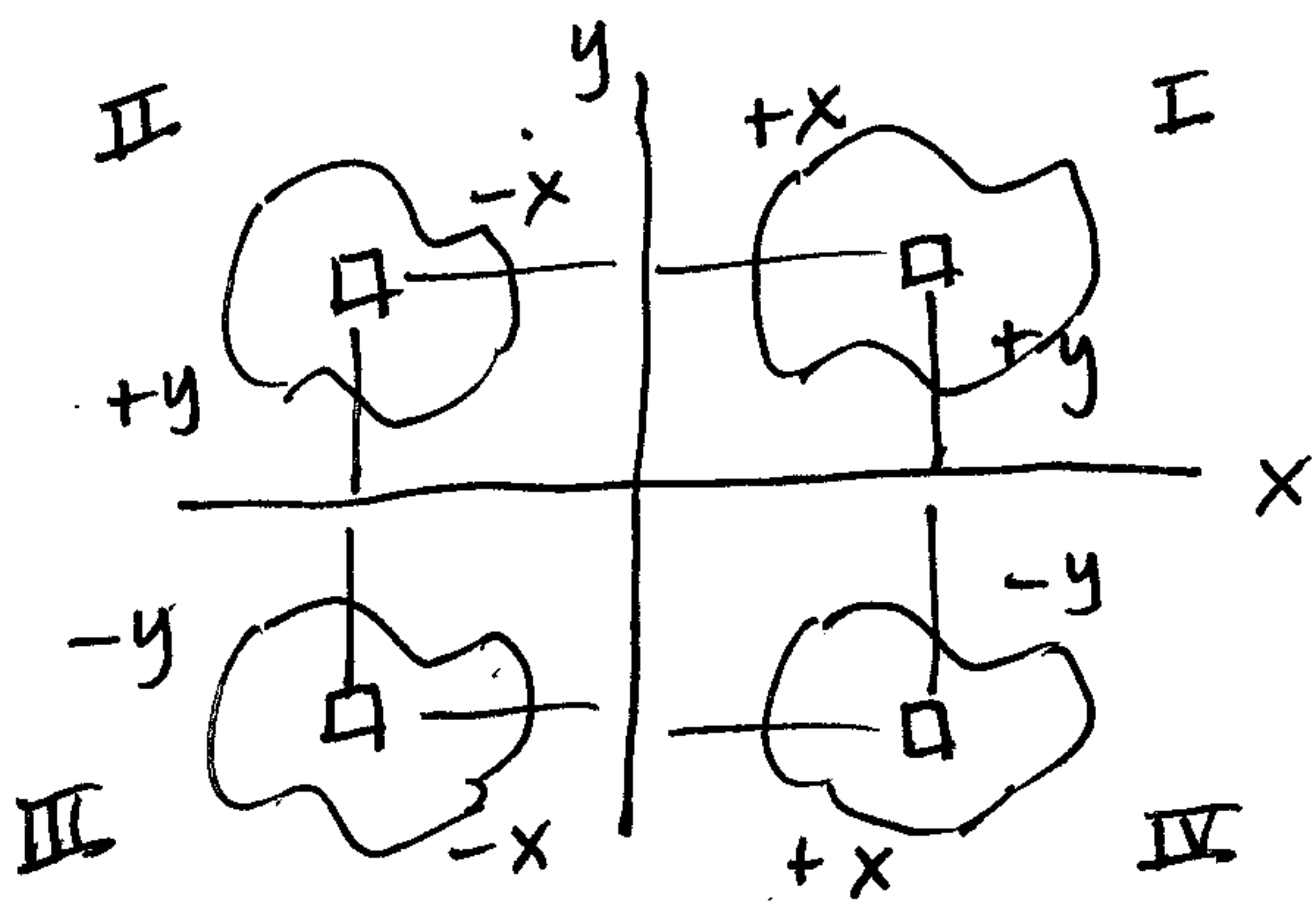


where half of the area is on each side, then
either the $+x, -x$ or/and $+y, -y$ dA terms will
cancel and

$$I_{xy} = \int_A xy dA = 0 \quad \square$$

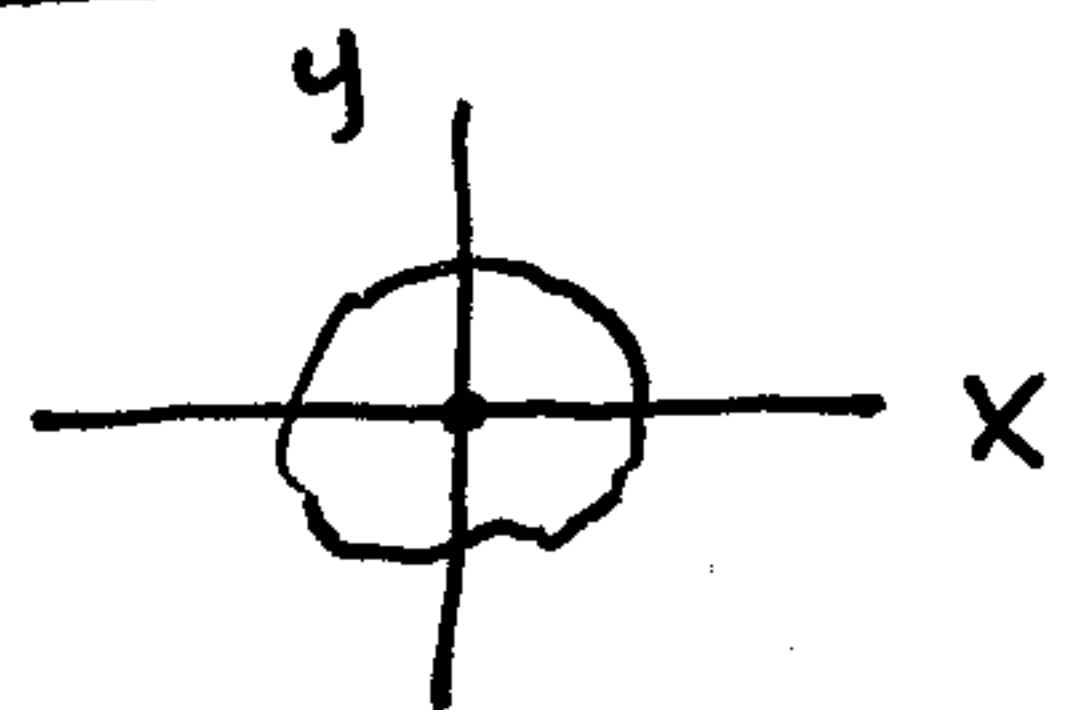
For areas in various quadrants ... the x

terms can be + or NEG,
and the y terms can be
+ or NEG.

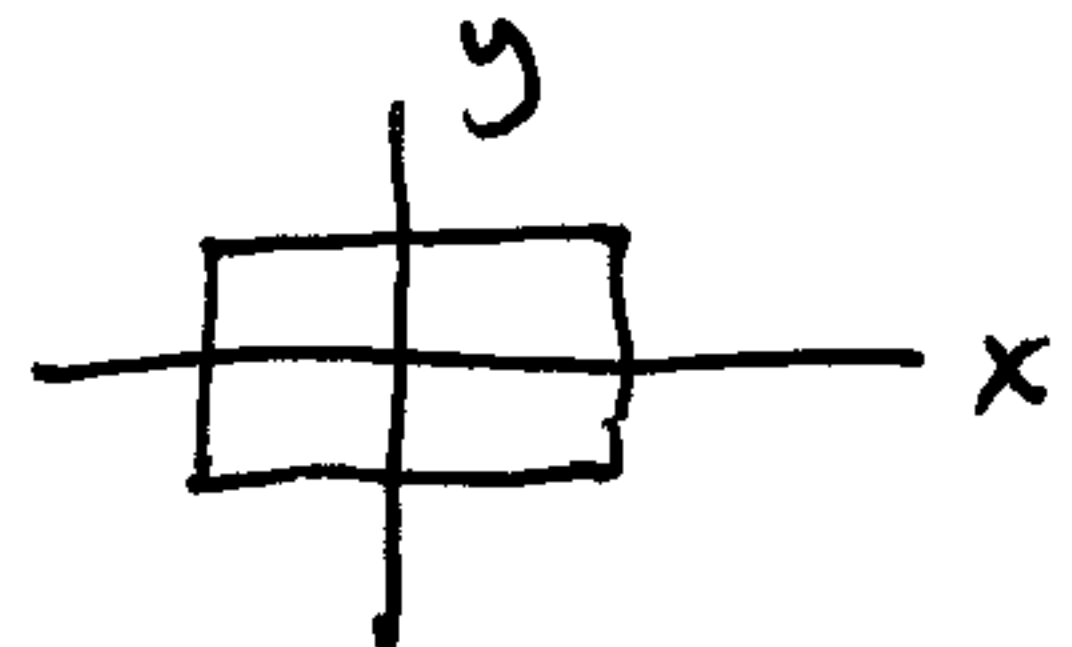


Quadrant	$(x)(y)$ Product	Sign of I_{xy}
I	$(+x)(+y) \rightarrow$	$+ I_{xy}$
II	$(-x)(+y) \rightarrow$	$- I_{xy}$
III	$(-x)(-y) \rightarrow$	$+ I_{xy}$
IV	$(+x)(-y) \rightarrow$	$- I_{xy}$

Exactly at
center
(Centroid)



$$I_{xy} = 0$$

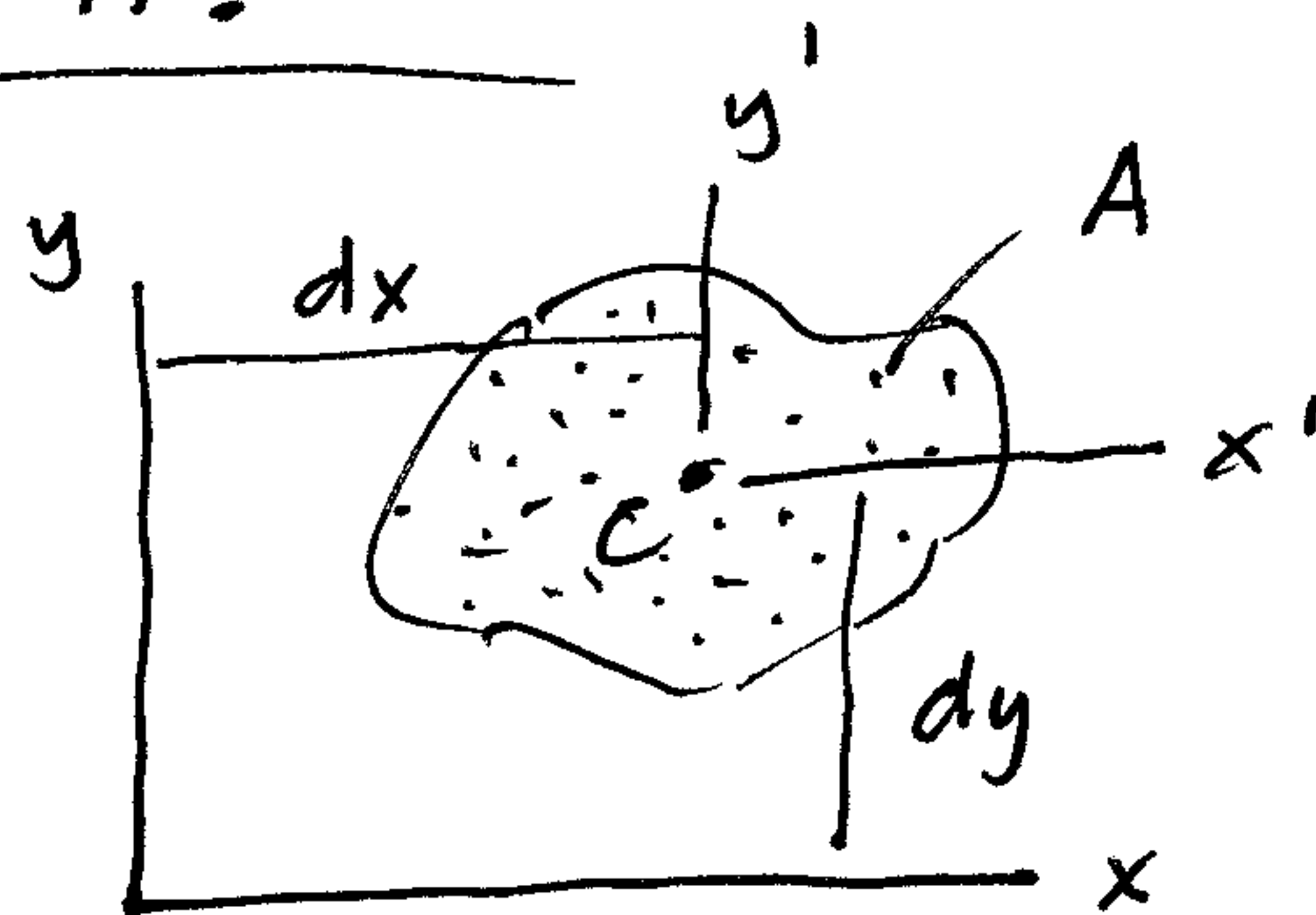


There is a Parallel Axis Theorem (PAT)
for I_{xy} . We will derive it.

③

$$I_{xy} = \bar{I}_{x'y'} + A dx dy$$

$\bar{I}_{x'y'}$ = POI about centroidal
axes (x', y') of shape



A = area

dx measured from y axis to y' axis.
 dy " " " x axis to x' axis. } dx and dy can be + or neg or zero.

I_{xy} = POI about $\begin{matrix} y \\ \text{---} \\ x \end{matrix}$ axes
(x parallel to x' , y parallel to y')

See shapes table for $\bar{I}_{x'y'}$ values.

Note: Some are zero! Some are +/-!

Is there an I_{base} for POI (as for I_x) or I_y ?

Generally, No. Remember:

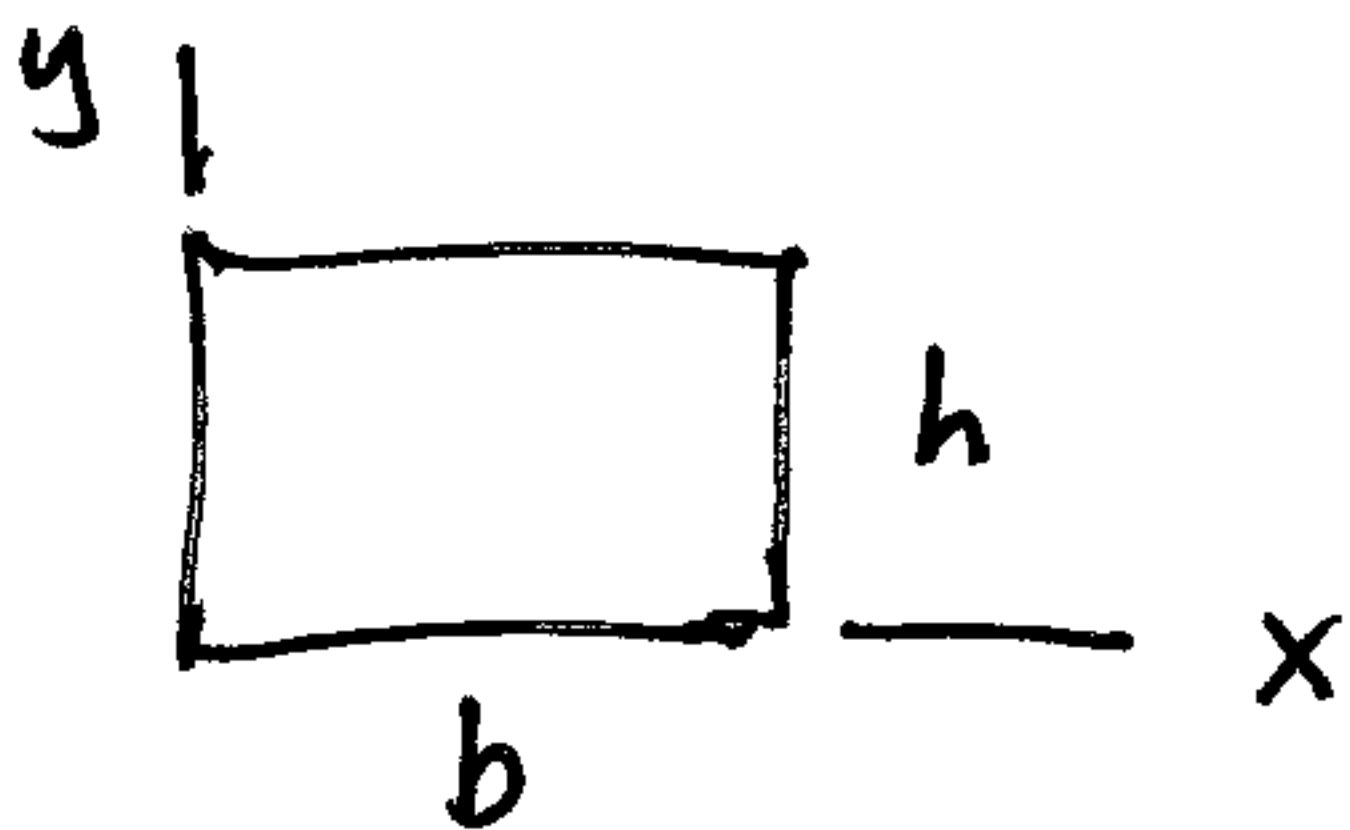
I_x and I_y are with respect
to a single axis,
so an I_{base} makes sense.

But, I_{xy} is with respect to a pair of axes
(usually I_{xy} , but also centroidal $\bar{I}_{x'y'}$).

So, even though there is no I_{base} for I_{xy} , ④
 some books and tables give what I call
 " I_{origin} " values for I_{xy} .

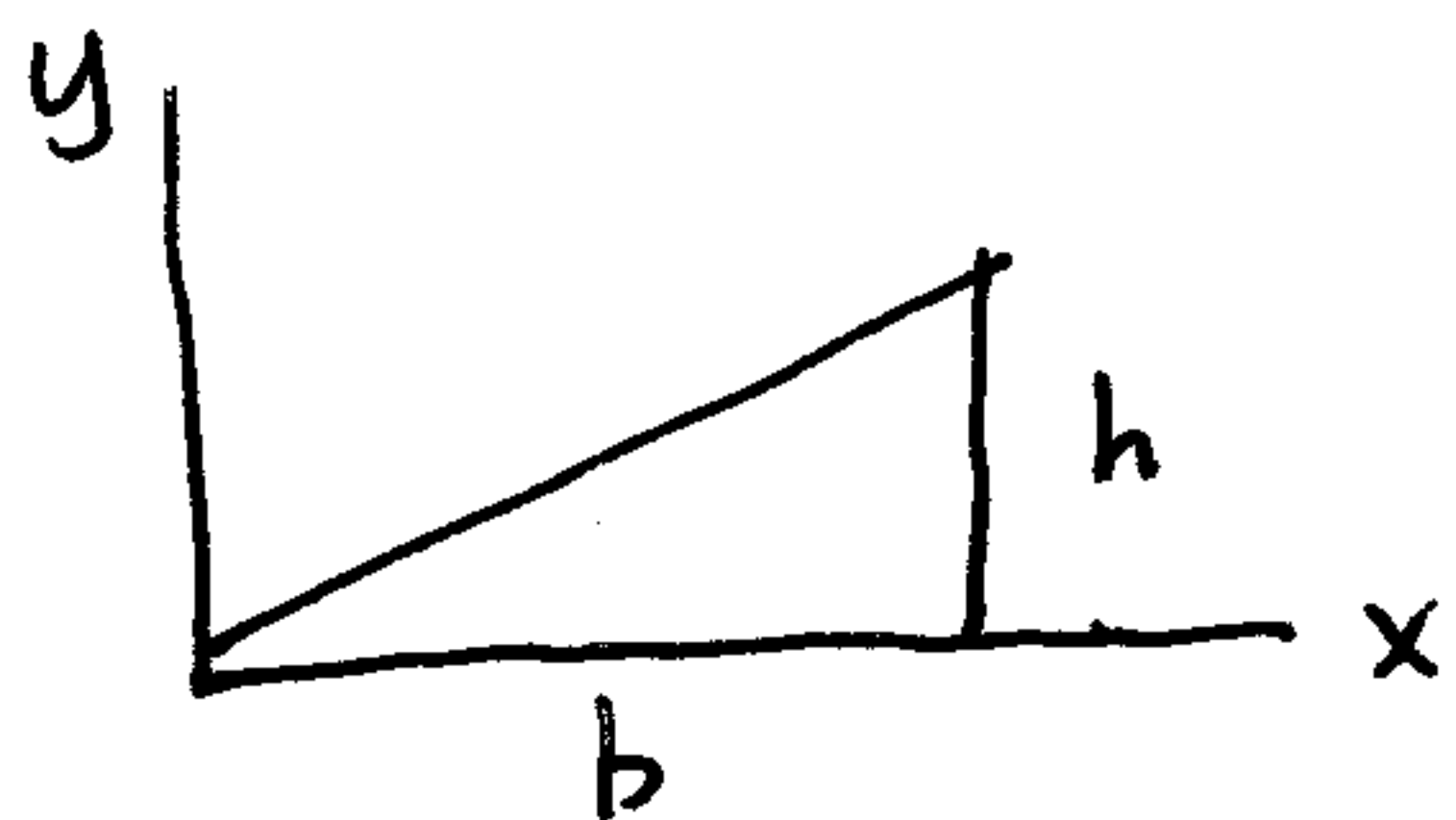
Hibbeler does this on his crib sheet.

He gives:



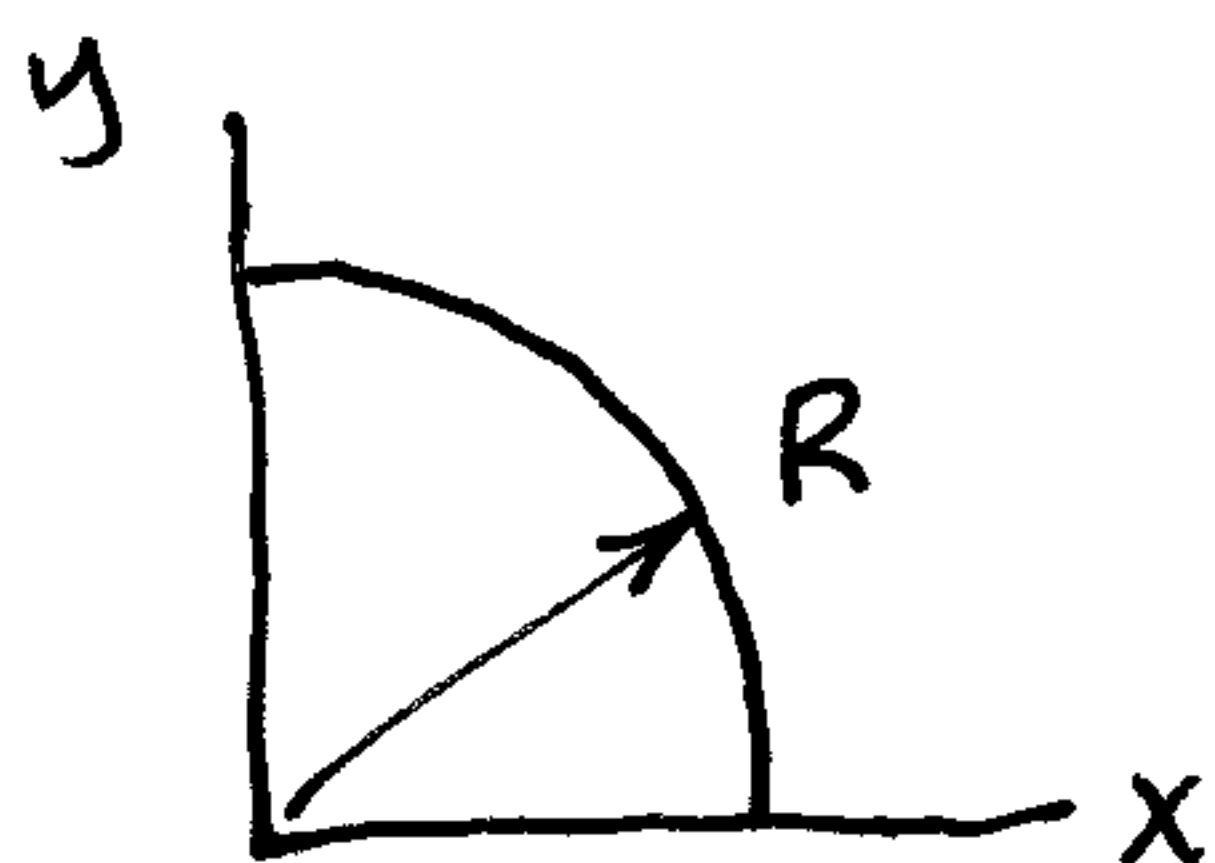
$$I_{xy} = \frac{1}{4} b^2 h^2$$

For a rectangle at
the origin.



$$I_{xy} = \frac{1}{8} b^2 h^2$$

For a triangle at this
 orientation at the origin.



$$I_{xy} = \frac{1}{8} R^4$$

for a quarter
 circle at the origin.

* This can be misleading! He does not indicate
 that these signs can vary! by quadrant.

Later I will give more complete " I_{origin} "
 formulas which give the proper signs, too.

For example:

-	+
+	-

$$I_{xy} = \pm \frac{1}{4} b^2 h^2$$

depending on the quadrant!

Kinds of POI (I_{xy}) Problems

⑤

We will work:

① I_{xy} by Integration.

② I_{xy} for Composite Areas

Then we will use I_{xy} (and I_x, I_y) to compute:

③ $I_{x'}, I_{y'}, I_{x'y'}$ for rotated axes θ from (x, y)

④ I_{max}, I_{min} at angles θ_{p1}, θ_{p2}

We will use transformation equations, such as:

$$I_{x'} = I_x \cos^2 \theta + I_y \sin^2 \theta - 2I_{xy} \sin \theta \cos \theta$$

or Mohr's Circle (MC):

